

9th Class Physics – Past Paper 2023

Punjab Board (Rawalpindi Board)

Group-I (Paper Code: 5471)

Objective Type

Marks: 12

Time: 15 Minutes

Multiple Choice Questions (MCQs)

Q.No.	Question	A	B	C	D
1	Measuring cylinder is used to measure	Mass	Area	Volume ✓	Level of liquid
2	Least count of Digital Vernier Callipers	0.1 mm	0.01 mm	0.001 mm ✓	1 mm
3	A change in position is called	Speed	Velocity	Displacement ✓	Distance
4	Inertia depends upon	Force	Net Force	Mass ✓	Velocity
5	Two equal unlike parallel forces with different lines of action produce	Torque	A Couple ✓	Equilibrium	Neutral Equilibrium
6	Earth's gravitational force vanishes at	6400 km	Infinity ✓	42300 km	1000 km
7	Value of g increases with	Increase in mass	Increase in altitude	Decrease in altitude ✓	None
8	Work done is zero when angle between force and displacement is	45°	60°	90° ✓	180°
9	Lightest substance among the following metals	Copper	Mercury	Aluminium ✓	Lead
10	Water freezes at	0°F	32°F ✓	-273 K	0 K

Q.No.	Question	A	B	C	D
11	In solids, heat is transferred by	Radiation	Conduction ✓	Convection	Absorption
12	Thermal conductivity of wood	0.08 Wm⁻¹K⁻¹ ✓	0.59 Wm ⁻¹ K ⁻¹	0.03 Wm ⁻¹ K ⁻¹	0.8 Wm ⁻¹ K ⁻¹

Subjective Type

Marks: 48

Time: 1 Hour 45 Minutes

Section – I

Question 2

Write short answers of any FIVE.

1. What is a measuring cylinder? Write its uses.
2. Define Atomic Physics.
3. How do we use a stopwatch?
4. Define random motion and write one example.
5. Define scalars and vectors.
6. Define uniform velocity.
7. Define momentum and write its SI unit.
8. Define inertia.

Question 3

Write short answers of any FIVE.

1. Define resultant of forces.
2. What is meant by a rigid body?
3. State the Principle of Moments.
4. Define gravitational field strength.
5. What is GPS?
6. Define artificial satellite.
7. Write the SI unit of power and define it.
8. Define efficiency.

Question 4

Write short answers of any FIVE.

1. Define density of a substance and write its formula and SI unit.
 2. Why does a wooden block float on water?
 3. State two consequences of thermal expansion.
 4. Define Celsius scale and Kelvin scale of temperature.
 5. What is a thermometer? Why is mercury preferred as a thermometric substance?
 6. Differentiate between conduction and convection.
 7. Write the names of the four faces of Leslie's Cube.
 8. Why does a cup of hot tea become cold after some time?
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Section – II

Attempt any TWO questions.

Question 5

(a)

State and explain Newton's Second Law of Motion.

(4 Marks)

(b)

A train starts from rest with an acceleration of 0.5 ms^{-2} .

Find its speed in km h^{-1} when it has moved through **100 m**.

(5 Marks)

Question 6

(a)

What is meant by kinetic energy? Derive its equation.

(4 Marks)

(b)

A force acts on a body making an angle of **30°** with the horizontal.

The horizontal component is **20 N**.

Find the magnitude of the force.

(5 Marks)

Question 7

(a)

How can conductors and non-conductors be used in houses? Explain.

(4 Marks)

(b)

Weight of a metal spoon in air = **0.48 N**

Weight of the spoon in water = **0.42 N**

Find its density.

(5 Marks)

End of Paper